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Ventilation

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Cases

Ventilators on the Fly

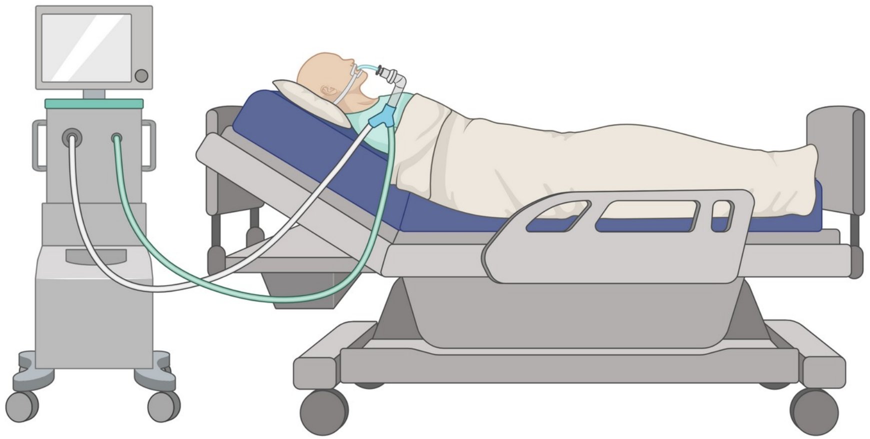
4 Basic Questions:

What is the mode?

What is the level of oxygen support?

Is ventilation appropriate?

How are the respiratory mechanics?



Authors: David Scudder, MD; Kaitlyn McLeod, MD; Arun Kannappan, MD
Section Editor: Trevor Steinbach, MD
Executive Editor: Yilin Zhang, MD

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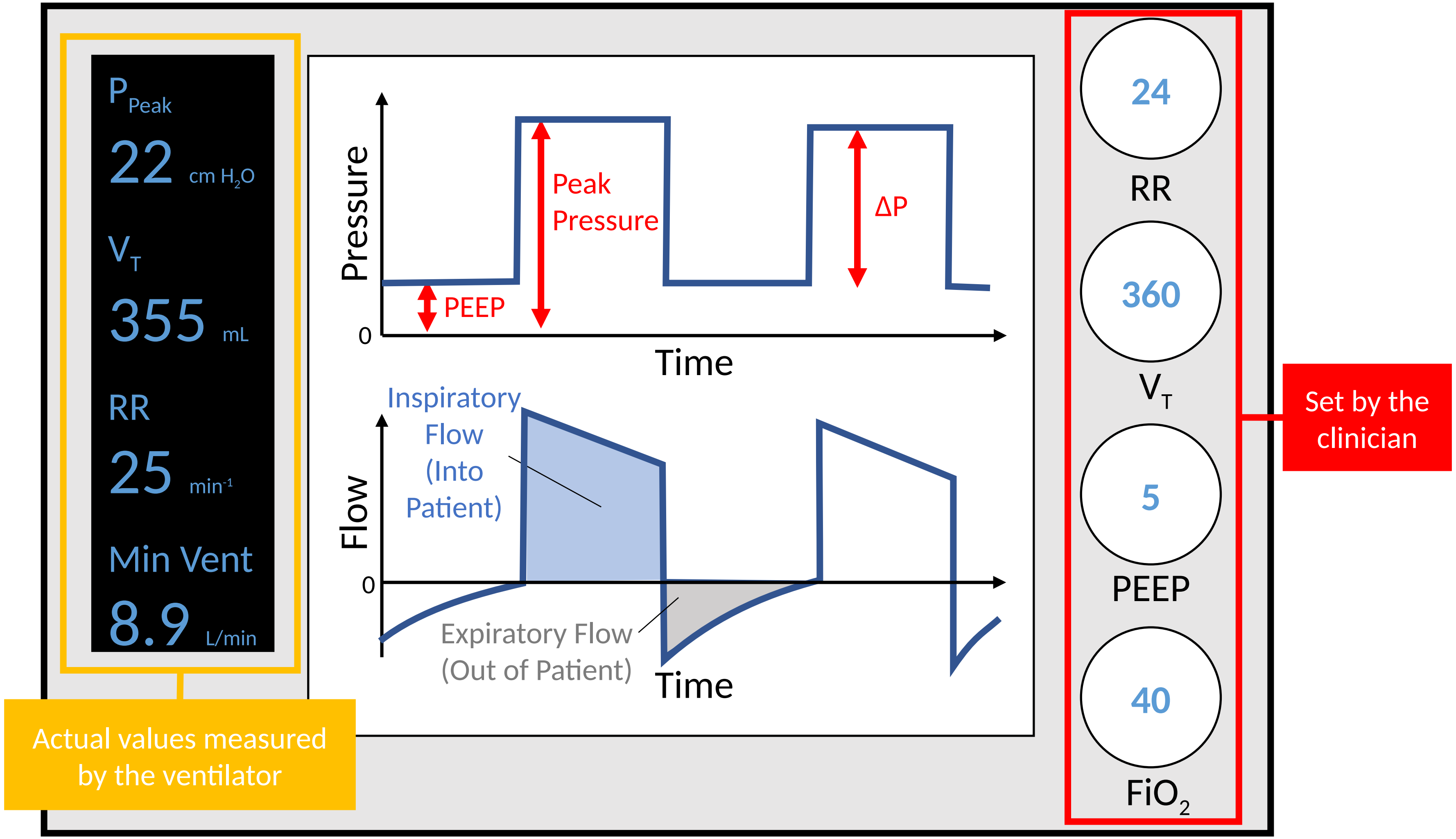
VENTILATOR ORIENTATION



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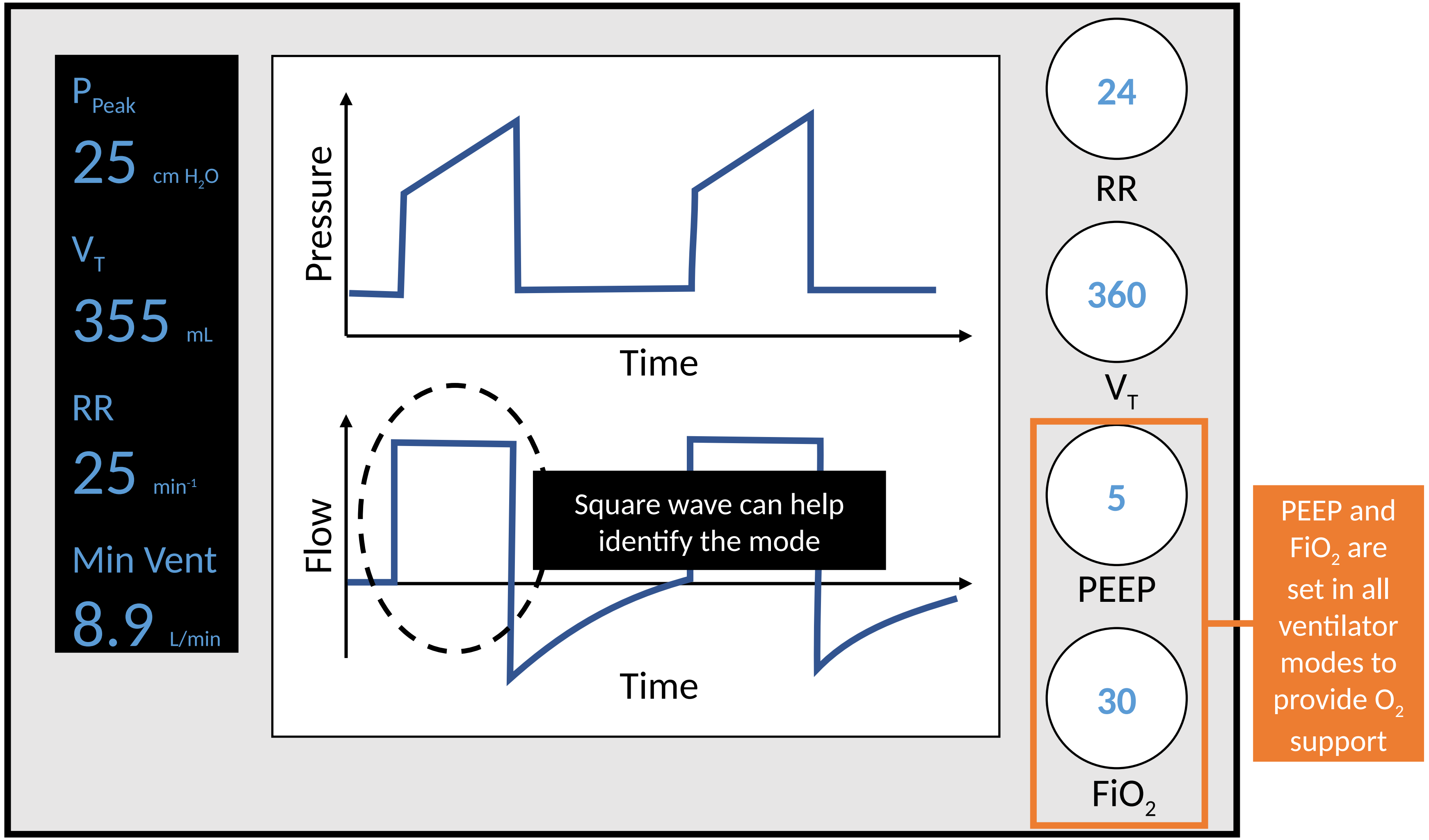
WHAT IS THE MODE?

Pressure Control | Volume Control | PRVC | Pressure Support



General Approach

1. Classic square wave
2. Mandatory RR?
3. What you target
4. What you monitor
5. When to use

Click for summary table of ventilator modes

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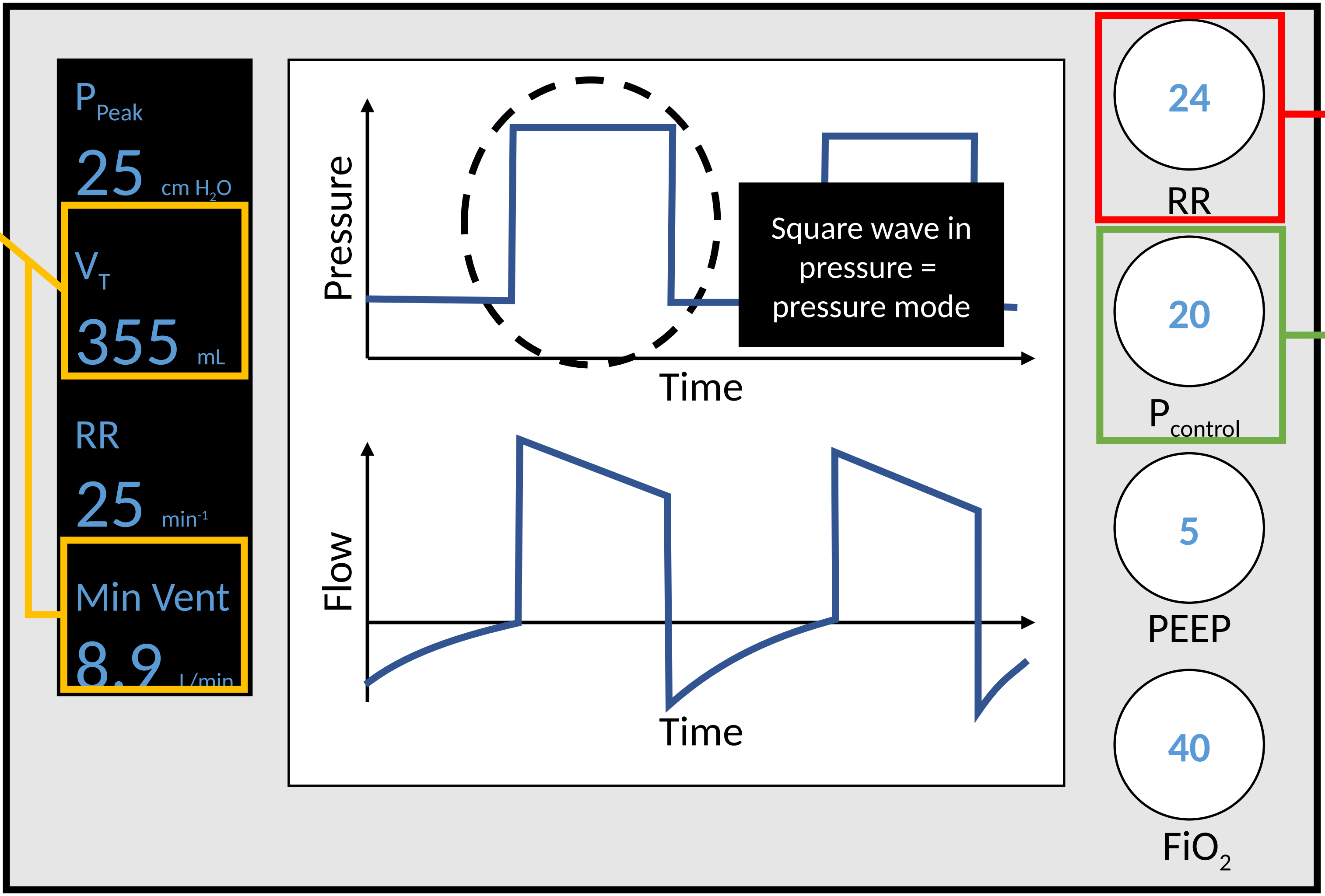
WHAT IS THE MODE?

Pressure Control | Volume Control | PRVC | Pressure Support

Key outputs are tidal volume and minute ventilation



"Control mode" = mandatory RR

Set target ΔP above PEEP

Pressure Control Mode

Classic square wave

Mandatory RR?

What you target

What you monitor

When to use

Patient comfort, precise airway pressure adjustment

Summary of ventilator modes

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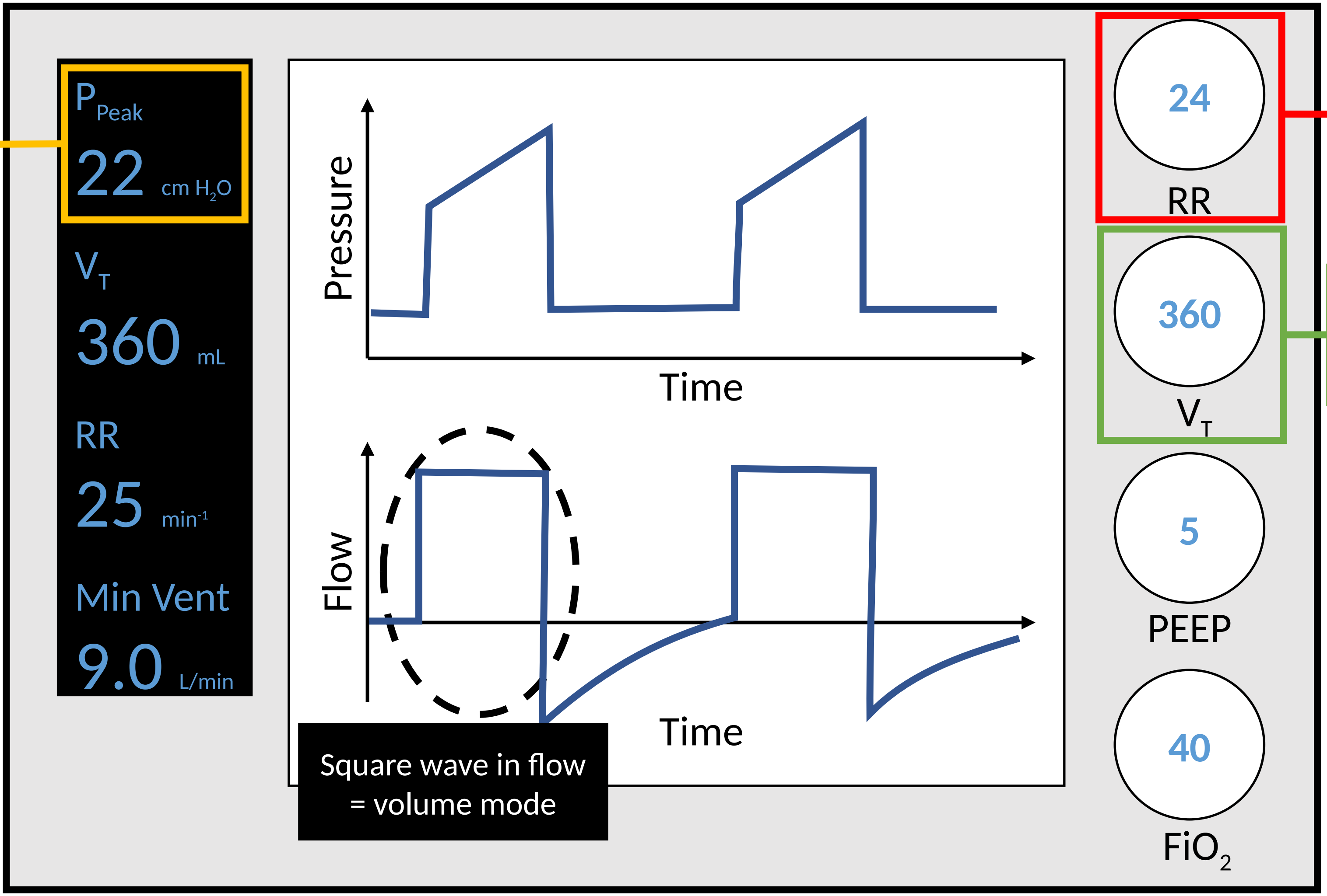
WHAT IS THE MODE?

Pressure Control | Volume Control | PRVC | Pressure Support

Key output is peak pressure



"Control mode" = mandatory RR

Set target tidal volume (V_T)

Volume Control Mode

Classic square wave

Mandatory RR?

What you target

What you monitor

When to use

ARDS (lung-protective tidal volumes)

Summary of ventilator modes

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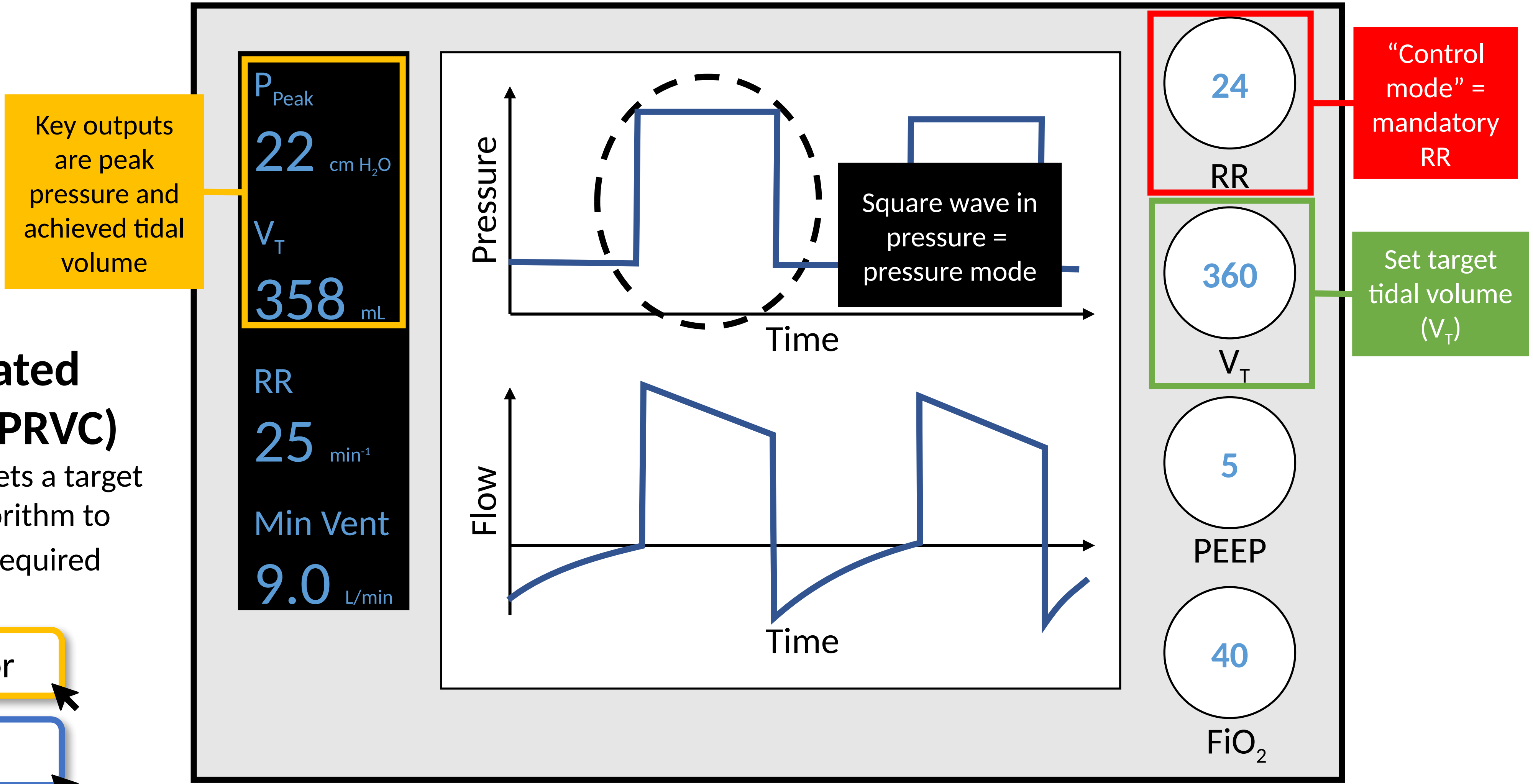
WHAT IS THE MODE?

Pressure Control | Volume Control | **PRVC** | Pressure Support



Pressure Regulated Volume Control (PRVC)

Combined mode. Clinician sets a target V_T. Ventilator uses an algorithm to determine the pressure required to achieve target.

What you monitor

When to use

Lung protective tidal volumes with comfort of pressure support

Summary of ventilator modes

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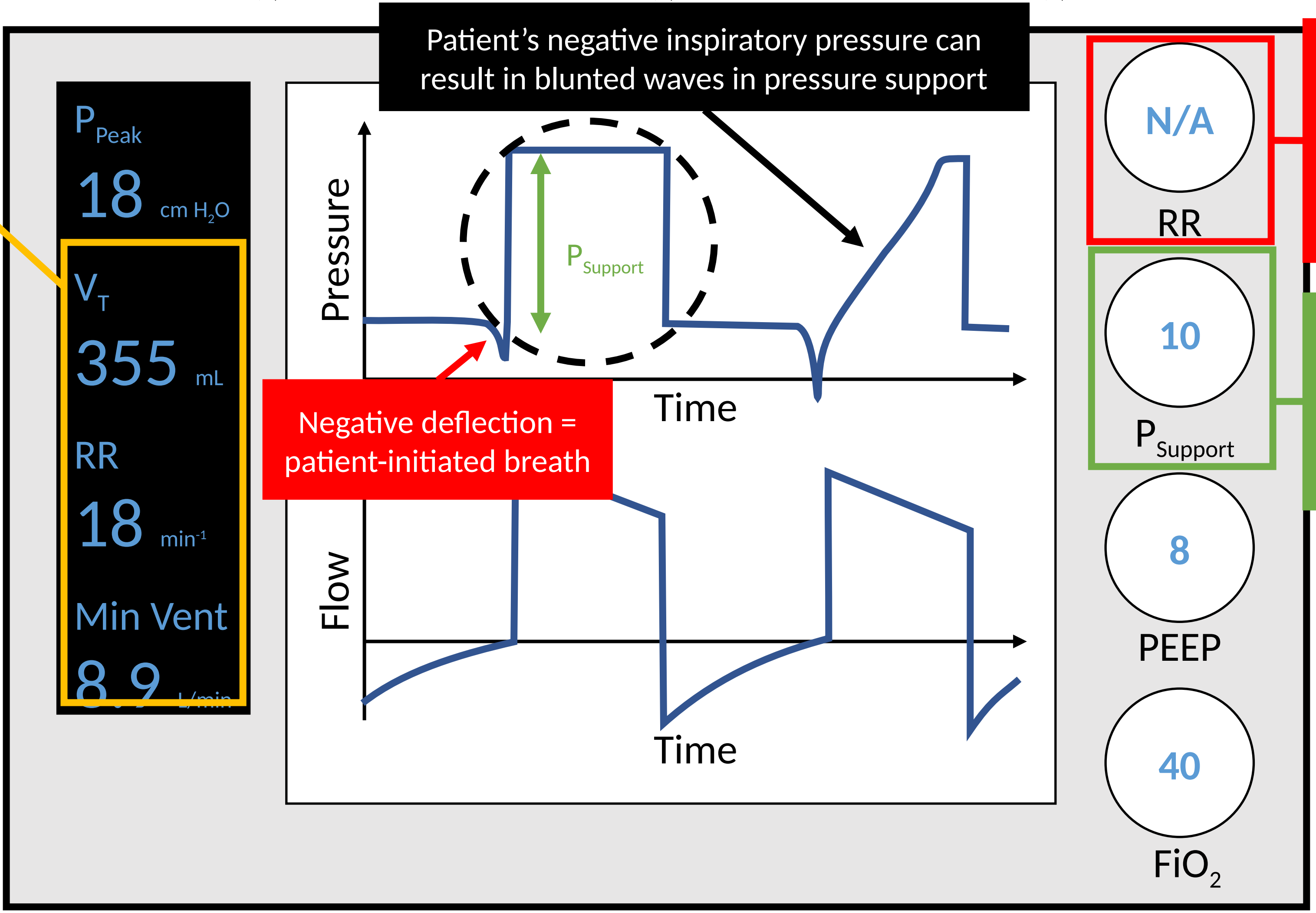
WHAT IS THE MODE?

Pressure Control | Volume Control | PRVC | **Pressure Support**

Key outputs are tidal volume, RR, and minute ventilation



"Support mode" = no mandatory RR. Patient initiates all breaths.

Set pressure support (positive pressure on inspiration)

Pressure Support Mode

Classic square wave

Mandatory RR?

What you target

What you monitor

When to use

SBTs, patient providing adequate RR

Summary of ventilator modes

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VENTILATOR MODE SUMMARY

Pressure Control | Volume Control | PRVC | Pressure Support

Mode	Classic square wave	Mandatory RR?	What you target	What you monitor	When to use
Pressure Control	Pressure	Yes	Inspiratory Pressure ($P_{Control}$)	Tidal Volume (V_T) and Minute Ventilation (V_E)	Patient comfort, precise airway pressure adjustment
Volume Control	Flow	Yes	Tidal Volume (V_T)	Peak Pressure and Plateau Pressure	ARDS (lung-protective tidal volumes)
Pressure Regulated Volume Control	Pressure	Yes	Tidal Volume (V_T)	Peak Pressure, Plateau Pressure, Achieved Tidal Volume (V_T)	Lung protective tidal volumes with comfort of pressure support
Pressure Support	Pressure	No	Pressure support ($P_{Support}$)	Tidal Volume (V_T), Respiratory Rate (RR), and Minute Ventilation (V_E)	SBTs, patient providing adequate RR

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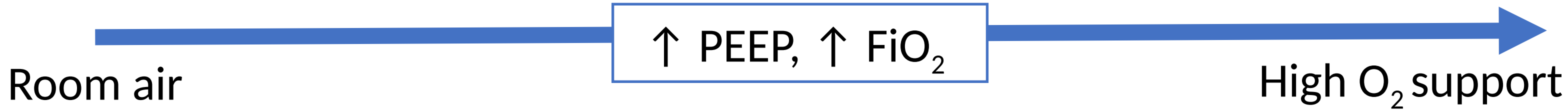
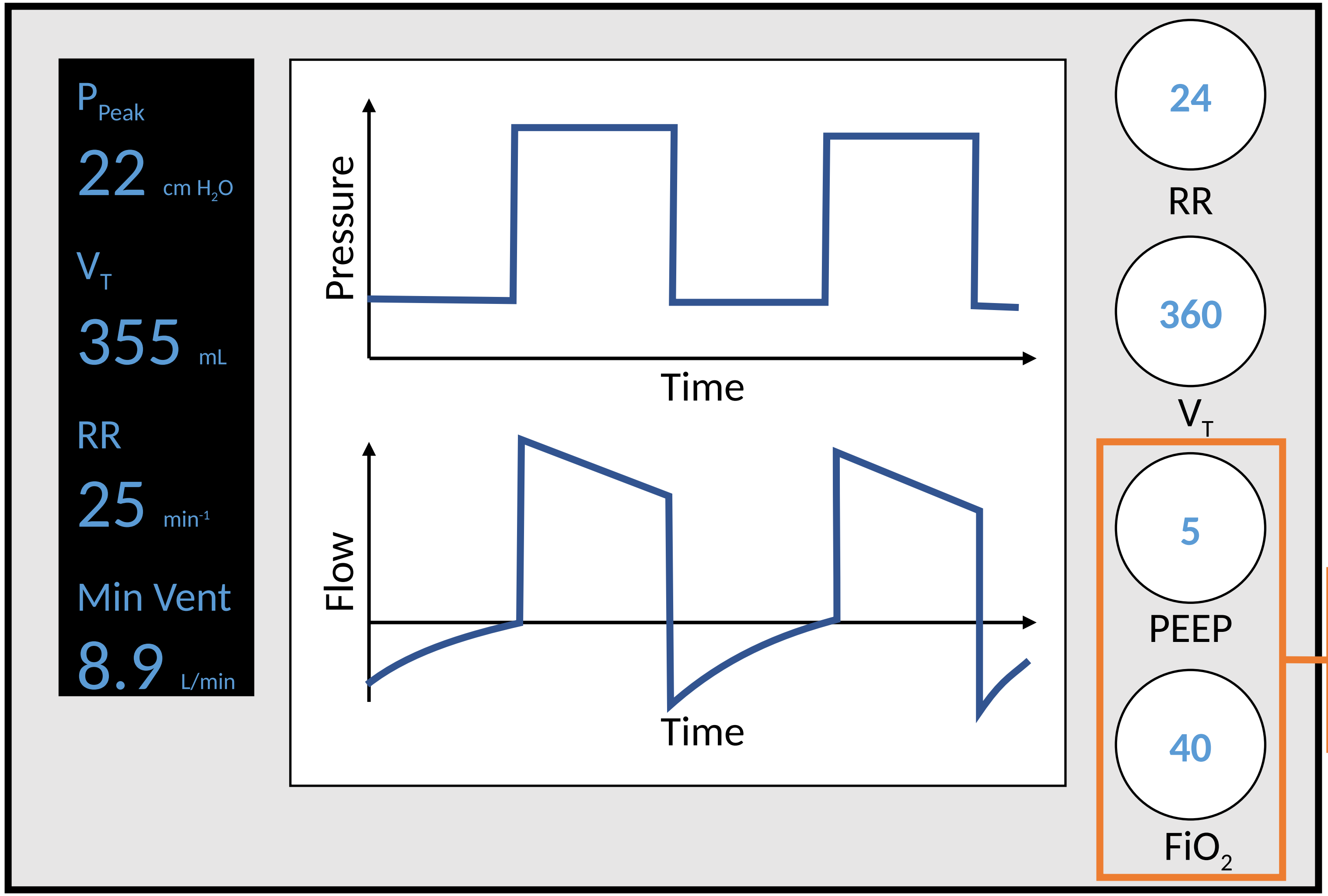
WHAT IS THE LEVEL OF O_2 SUPPORT?



FiO_2	PEEP
30%	5
50%	8
70%	10
100%	18-24

The ARDSNet PEEP ladder gives a sense of where the patient is with oxygen support.



Higher PEEP and FiO_2 = higher O_2 support

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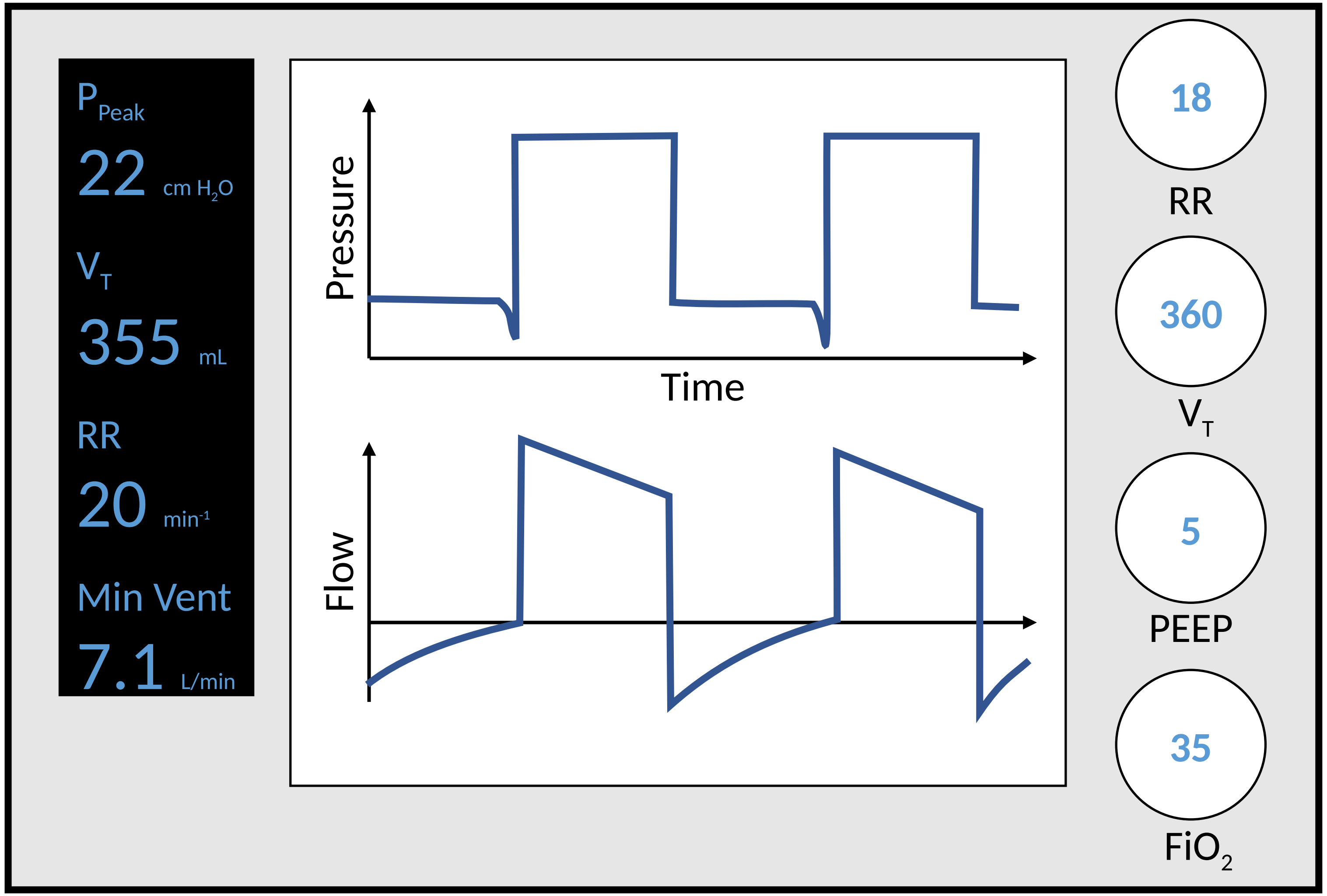
IS VENTILATION APPROPRIATE?

1. Check the minute ventilation

2. Is it appropriate?

3. Vent or Patient?



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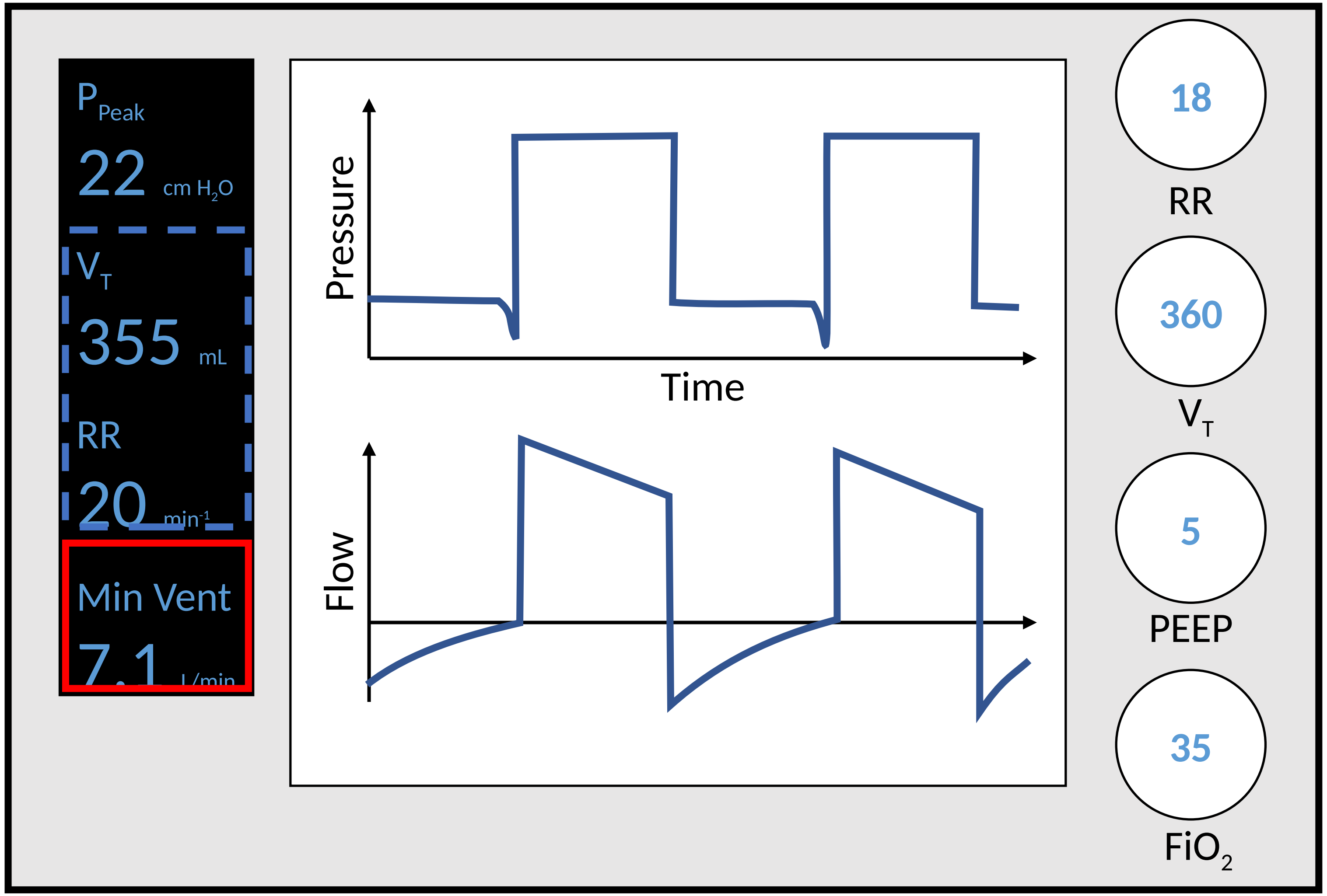
IS VENTILATION APPROPRIATE?

1. Check the minute ventilation

2. Is it appropriate?

3. Vent or Patient?



Min Vent (V_E) = V_T x RR

7.1 L/min = 0.355 L x 20 min^{-1}

Normal is 4-6 L/min
Often much higher when disease is present

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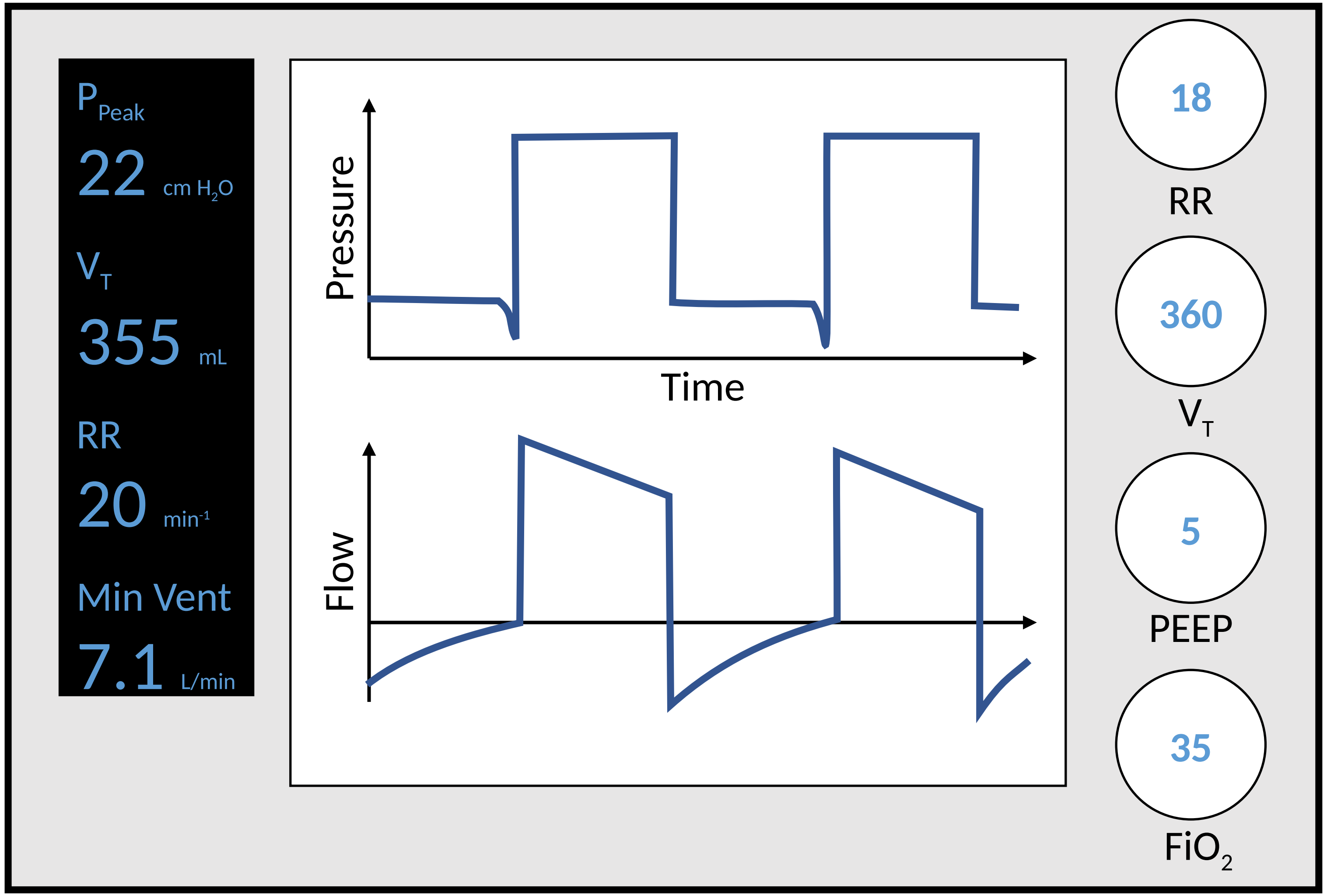
IS VENTILATION APPROPRIATE?

1. Check the minute ventilation

2. Is it appropriate?

3. Vent or Patient?



$\boxed{pH}/pCO_2/pO_2/HCO_3$

General rule is to maintain pH 7.2-7.5

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IS VENTILATION APPROPRIATE?

1. Check the minute ventilation

2. Is it appropriate?

3. Vent or Patient?

Is patient triggering breaths?

no

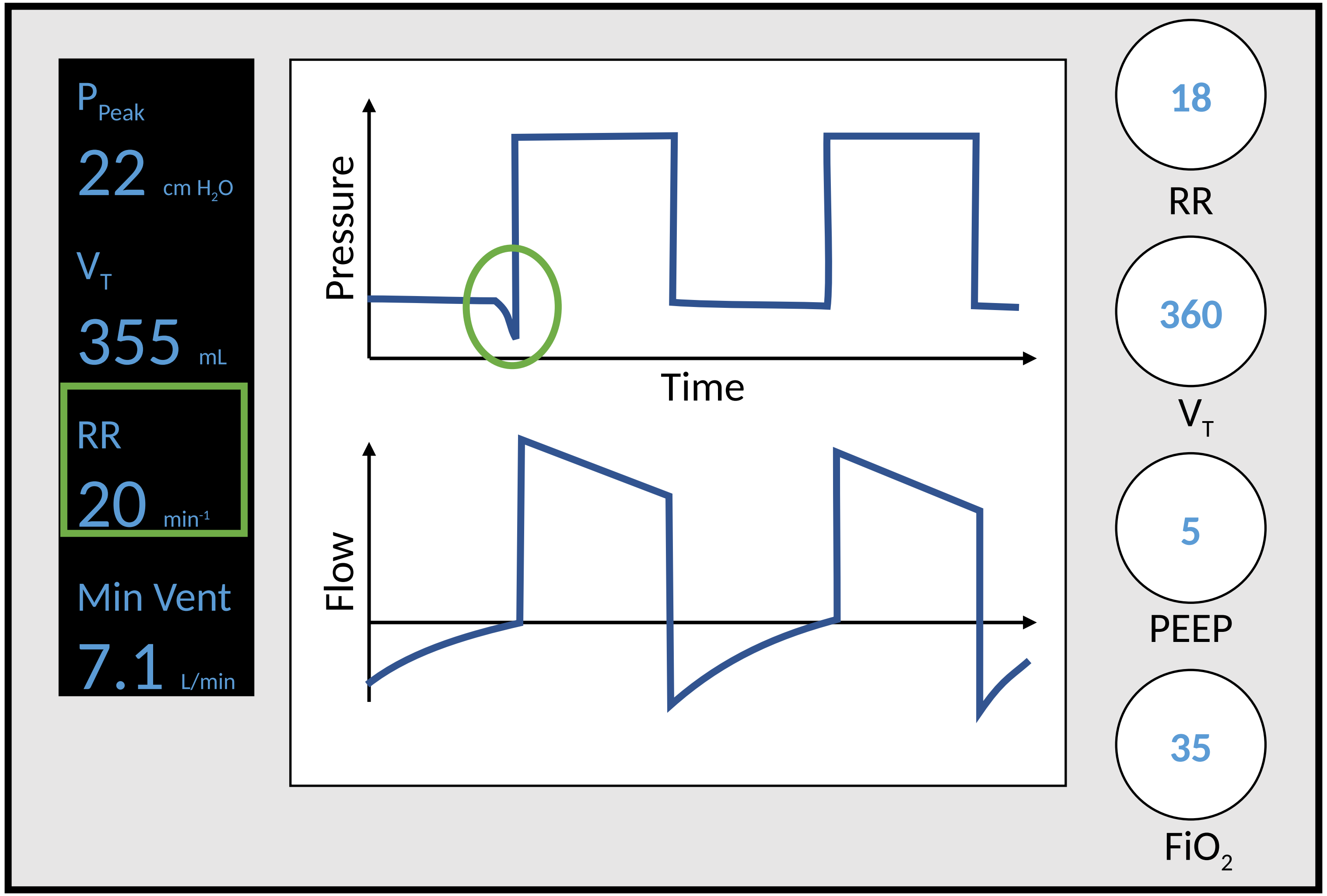
Adjust ventilator settings

$V_E = V_T \times RR$

yes

Look for patient causes

CNS suppression
CNS injury
Pain
Salicylates



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HOW ARE THE RESPIRATORY MECHANICS?

Pressure-Targeted Modes

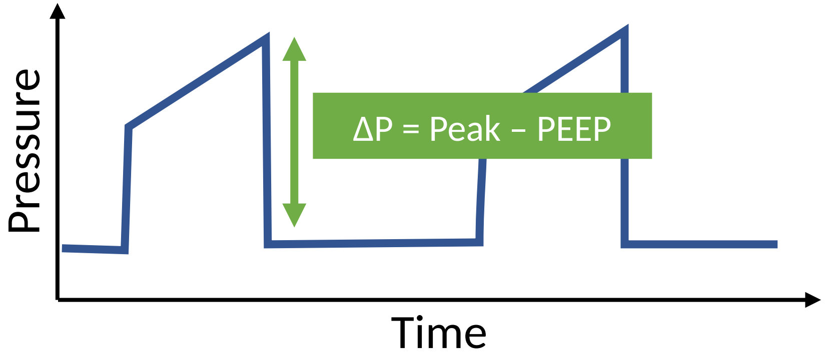
Key Parameters: V_T and V_E

Low V_T and V_E point to trouble with respiratory mechanics

Volume-Targeted Modes

Key Parameter: ΔP

Most lungs with good mechanics require a ΔP of **<10 cm H_2O** to achieve a 6 mL/kg tidal volume



What if ΔP > 10?

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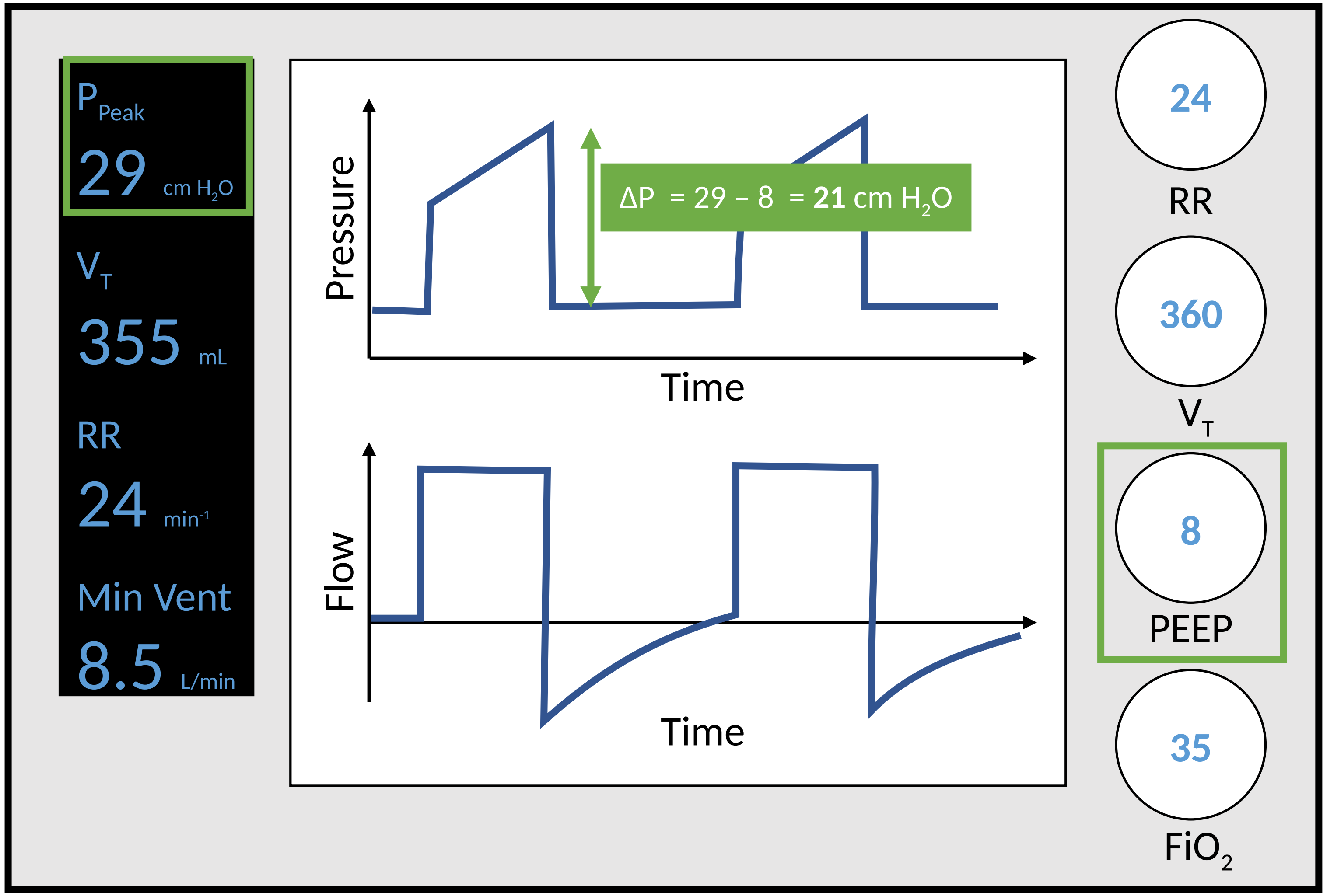
HOW ARE THE RESPIRATORY MECHANICS?

In Volume-targeted modes, if ΔP > 10, perform an inspiratory hold



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HOW ARE THE RESPIRATORY MECHANICS?

In Volume-targeted modes, if ΔP > 10, perform an inspiratory hold

Inspiratory hold

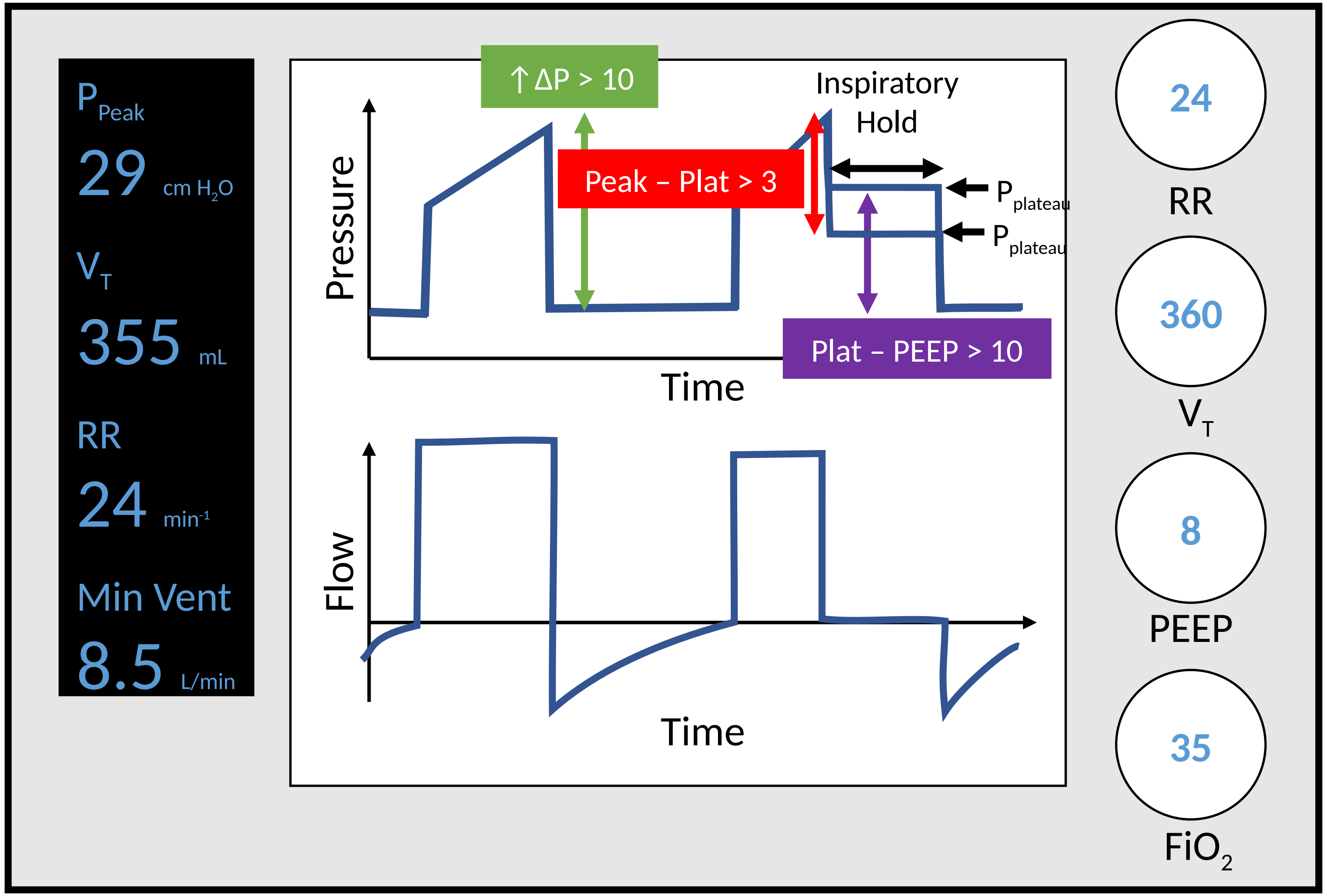
$P_{Plateau}$

Resistance Issue

Peak – Plat > 3
Mucous plug
Kinked tube
Bronchospasm

Compliance Issue

Plat – PEEP > 10
Pneumothorax
Alveolar filling
Fibrosis
Mainstem intubation
Distended abdomen



Case 1

What is the mode?

Pressure control

What is target parameter?

$P_{Control}$

What are the key outputs?

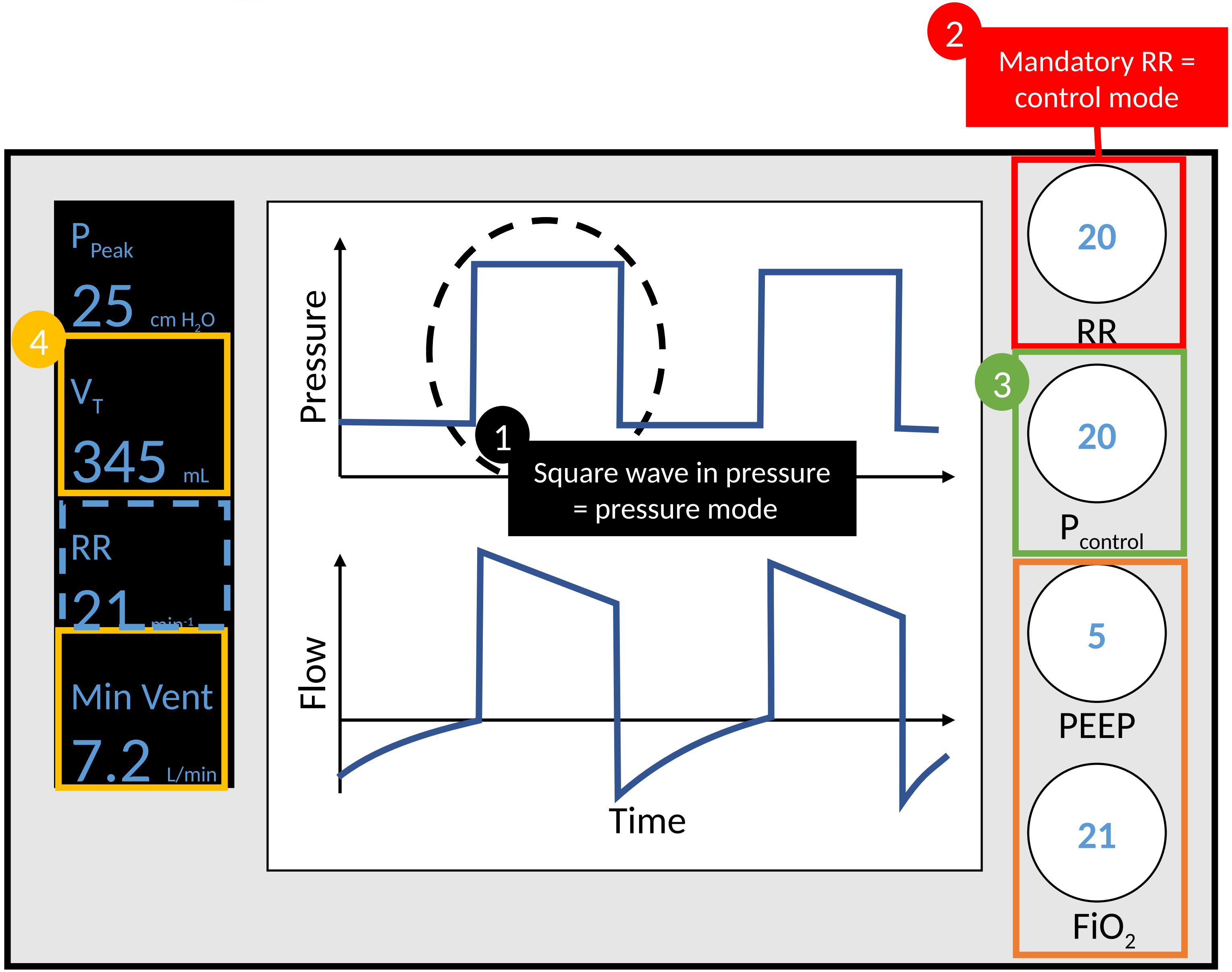
V_T and minute ventilation

What is the actual RR?

21 breaths per minute

How much O_2 support is the patient on?

Low O_2 support – patient is on minimal FiO_2 and PEEP



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Case 2

What is the mode?

Volume control

What is target parameter?

V_T

What are the key outputs?

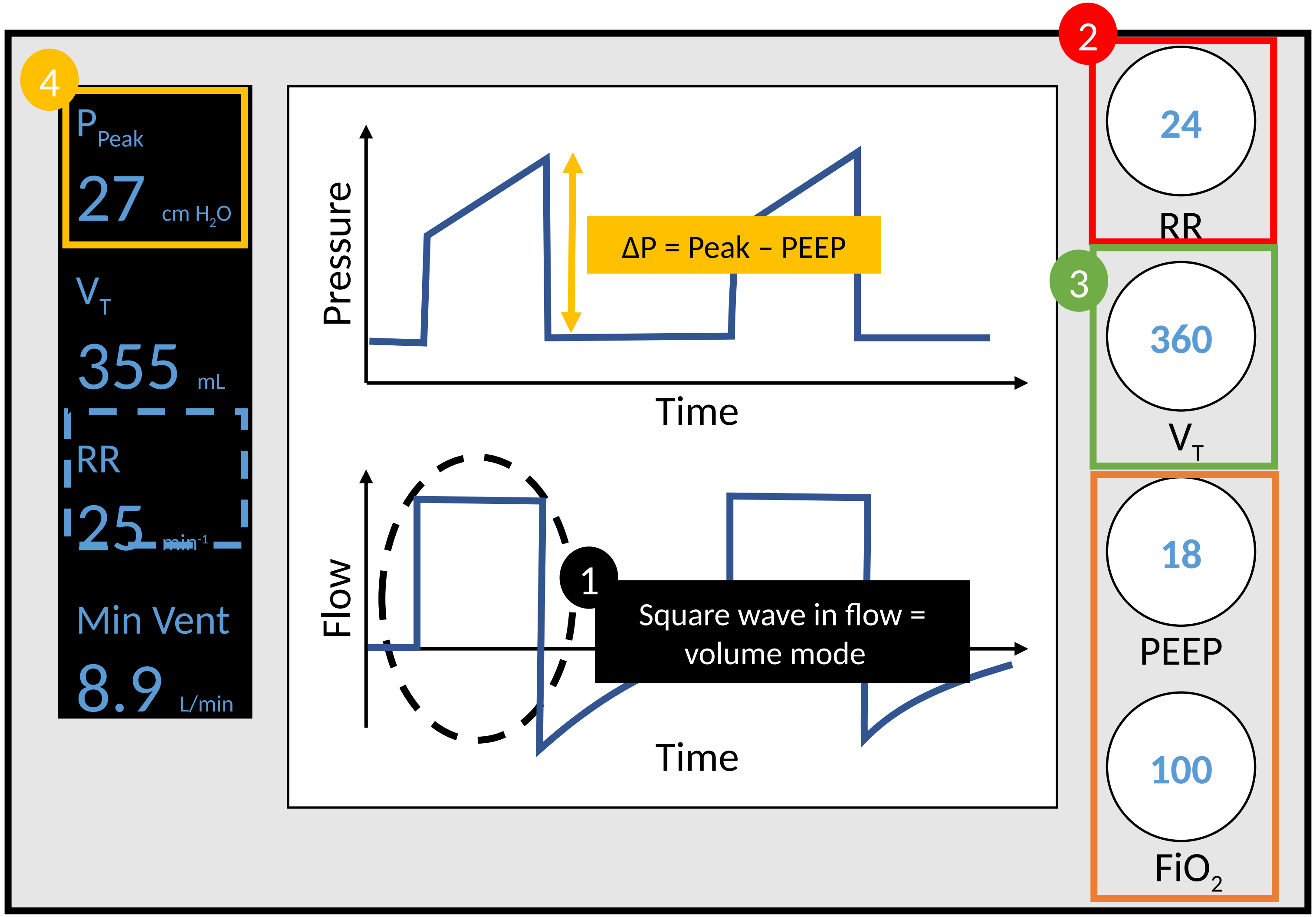
Peak pressure ($\Delta P < 10$, so inspiratory hold not needed)

What is the actual RR?

25 breaths per minute

How much O_2 support is the patient on?

Very high O_2 support.
Maximum FiO_2 and PEEP.



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Case 3

What is the mode?

Pressure support

What is target parameter?

$P_{Support}$

What are the key outputs?

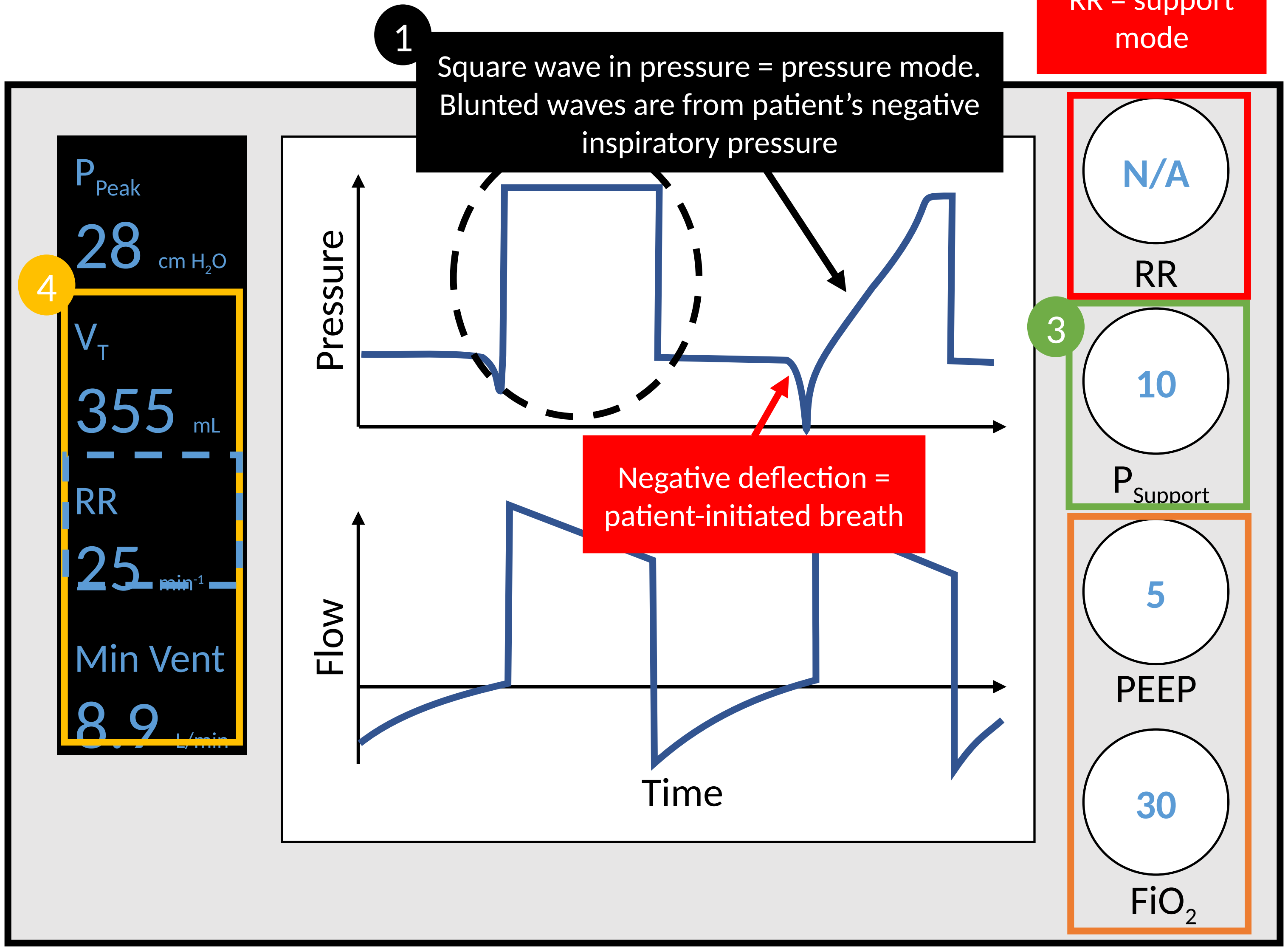
V_T, RR, and minute ventilation

What is the actual RR?

25 breaths per minute

How much O_2 support is the patient on?

Low O_2 support. Low FiO_2 and PEEP



Modern Framing — Lung-Protective Ventilation (unchanged core, 2015–24 evidence)

The original talk teaches mode identification; this slide adds current ARDS ventilation targets. The standing teaching is unchanged.

Low tidal volume (still the foundation)

• 6 mL/kg predicted body weight (range 4–8) in ARDS; the original ARDSNet mortality benefit (NEJM 2000) remains standard of care.

Pressure limits

• Keep plateau pressure <30 cm H_2O (many target ≤28).

Driving pressure (ΔP = Plateau − PEEP) — the newer prognostic lever

• Lower driving pressure is the ventilator variable most strongly associated with survival in ARDS (Amato, NEJM 2015; mediation analysis of 3,562 patients). Target ΔP ≤14–15 cm H_2O where achievable.

• Note: on this slide ΔP = Peak − PEEP (a quick bedside screen). True driving pressure uses the PLATEAU pressure (inspiratory hold), not peak.

Sources: ARDSNet, NEJM 2000;342:1301. Amato MBP et al, NEJM 2015;372:747. Lung-protective targets unchanged through 2024.

Take-Home Points

- Read any ventilator with 4 questions: (1) What mode? (2) What O_2 support? (3) Is ventilation appropriate? (4) How are the mechanics?
- Mode clues: square wave in PRESSURE = pressure mode (PC, PRVC, PS); square wave in FLOW = classic volume control. A set/mandatory RR = control mode; no mandatory RR = support mode (PS).
- Target vs monitor: set pressure → monitor VT & minute ventilation; set volume → monitor peak & plateau pressure. PEEP and FiO_2 set the O_2 support in every mode (ARDSNet PEEP/FiO_2 ladder).
- Ventilation: VE = VT × RR (normal 4–6 L/min, often higher in disease); aim to keep pH ~7.2–7.5. Check whether breaths are vent- or patient-triggered.
- Mechanics: in volume-targeted modes, if $\Delta P > 10$ do an inspiratory hold. Peak − Plateau > 3–5 = RESISTANCE problem (mucus plug, kinked tube, bronchospasm); elevated Plateau − PEEP (>10) = COMPLIANCE problem (pneumothorax, edema/alveolar filling, fibrosis, mainstem intubation, distended abdomen).

Attribution & Changelog

Original

• "Ventilators on the Fly" — Scudder D, McLeod K, Kannappan A; sec. ed. Steinbach T; exec. ed. Zhang Y. TeachIM.org, Dec 2021. https://teachim.org/teaching_material/ventilators-on-the-fly/ — Educational use only; not medical advice.

Correction made to this copy (June 2026)

• Slide 16: "Compliance Issue" box label "Plat – Peak > 10" → "Plat – PEEP > 10" — fixes an internal inconsistency with the same slide’s waveform annotation; compliance is flagged by elevated plateau relative to PEEP.

Appended (originals untouched)

• Lung-protective / driving-pressure framing slide; take-home points; this attribution slide. All interactive animations and click-to-reveal builds preserved.

Companion files

• Learner Handout (PDF) and Teachers Guide (DOCX) copied unchanged into the lesson folder & linked — reviewed, no factual errors.